

PROJECT TITAN

Material Trade Study

Educational Engineering Project

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IMPORTANT SCOPE NOTICE

Project TITAN is an educational design and analysis project. It is not a clinically validated device, regulatory submission, or product cleared for human use.

Document Control and Revision History

Revision	Date	Description	Prepared By	Status
A	July 2026	Initial material screening and weighted selection for baseline structural components.	Michael Lozada-Longog	Released

Source baseline: Project TITAN Engineering Design Report (July 2026), CAD assembly, completed engineering drawings, and Fusion 360 static FEA results documented by the author.

1. Purpose

This trade study compares candidate materials for Project TITAN structural plates and frame members. The goal is not to identify a universally superior material, but to select the most appropriate baseline material for an educational, low-volume, CNC-oriented mechanical prototype.

2. Candidate Materials

- 6061-T6 aluminum
- 7075-T6 aluminum
- 316L stainless steel
- Ti-6Al-4V titanium alloy
- Carbon-fiber-reinforced polymer (CFRP) concept
- PEEK engineering polymer

3. Selection Criteria and Weighting

Criterion	Weight	Reason
Specific strength / mass efficiency	25%	Wearable-oriented concepts benefit from reduced mass while preserving strength.
Machinability and prototype simplicity	20%	Current development strategy relies on conventional CNC milling and turning.
Cost and availability	20%	Educational prototyping requires readily available stock and manageable replacement cost.
Corrosion/environmental resistance	15%	The concept may be exposed to humidity and handling; future wearable use could introduce sweat.
Stiffness and dimensional stability	10%	Alignment of the shaft and bearings depends on structural stiffness.
Documentation and supply familiarity	10%	Well-characterized, standardized materials simplify analysis and sourcing.

Scoring scale: 1 = poor, 2 = below average, 3 = acceptable, 4 = good, 5 = excellent. Scores represent an educational engineering judgment for this project scope, not certified supplier ratings.

4. Qualitative Property Comparison

Material	Relative Density	Relative Strength	Machinability	Corrosion Resistance	Prototype Cost	Key Tradeoff
6061-T6 Al	Low	Moderate	Excellent	Good	Low	Balanced baseline; lower strength than 7075 or titanium
7075-T6 Al	Low	High	Good	Moderate	Moderate	Higher strength, but costlier and less corrosion/weld friendly
316L Stainless	High	Moderate	Fair	Excellent	Moderate	Excellent corrosion resistance with substantial mass penalty
Ti-6Al-4V	Medium	Very high	Difficult	Excellent	Very high	Exceptional performance but expensive and difficult to machine
CFRP concept	Very low	Direction-dependent high	Specialized	Excellent	High	Anisotropy, joining, edge damage, and inspection complexity
PEEK	Low	Low to moderate	Good	Excellent	High	Light and chemically resistant but low stiffness for baseline structure

5. Weighted Decision Matrix

Material	Specific strength	Machinability	Cost/availability	Corrosion resistance	Stiffness	Supply familiarity	Weighted Score
6061-T6 Al	4	5	5	4	3	5	4.40/5
7075-T6 Al	5	4	3	3	3	4	3.80/5
316L Stainless	2	2	3	5	5	5	3.25/5
Ti-6Al-4V	5	1	1	5	4	3	3.10/5
CFRP concept	5	1	1	5	4	2	3.00/5
PEEK	2	4	2	5	1	3	2.85/5

6. Selection Rationale: 6061-T6 Aluminum

6.1 Strength-to-Weight Balance

6061-T6 provides a practical combination of low density, moderate yield strength, and sufficient stiffness for the current load case. The refined assembly FEA met the project safety-factor target with a reported minimum safety factor of 2.377 under the documented idealized setup. Therefore, the baseline does not presently require the cost and manufacturing burden of a higher-performance alloy solely to satisfy the educational static requirement.

6.2 Manufacturability

The geometry is dominated by plates, bores, slots, spacers, and a cylindrical shaft. These features align well with CNC milling, drilling, and turning in 6061-T6. The alloy is familiar to machine shops, commonly stocked, and suitable for iterative prototype changes.

6.3 Environmental Suitability

Aluminum develops a passive oxide layer and can be anodized for improved surface durability. This makes 6061-T6 preferable to unprotected carbon steel for humid handling environments. This statement does not establish sweat, cleaning-agent, or biocompatibility performance; those conditions require separate verification.

6.4 Cost and Educational Value

6061-T6 allows the design to be fabricated and, if desired, destructively tested without the high material and machining costs associated with titanium or composite construction. It is therefore a strong baseline for learning, iteration, and physical validation.

7. Part-Specific Material Recommendations

Component	Baseline	Future Candidate	Reason for Future Review
Side plates	6061-T6 aluminum	7075-T6 or topology-optimized 6061	Potential mass reduction or higher fatigue margin
Upper/lower frames	6061-T6 aluminum	6061-T6 retained or CFRP hybrid	Mass and ergonomic improvement
Spacers	6061-T6 aluminum	Acetal/PEEK depending interface	Friction, wear, isolation, and mass
Main shaft	6061-T6 in current drawings	17-4PH stainless or hardened alloy steel	Bearing wear, surface hardness, bending, and durability
Bearings	52100 bearing steel concept	Sealed corrosion-resistant COTS bearing	Contamination and wearable-environment durability

Important Design Note

The current drawing package identifies the main shaft as 6061-T6 aluminum. For a physical, cyclic, or wearable prototype, the shaft material must be revisited because bearing contact, wear resistance, surface hardness, and repeated bending may make a steel shaft preferable.

8. Sensitivity and Decision Risks

- If measured loads exceed the current static assumptions, the material ranking may shift toward 7075-T6 or titanium.
- If corrosion and cleaning compatibility dominate, 316L stainless may become more attractive despite its mass.
- If mass becomes the dominant requirement, CFRP may rank higher but would require a new joint, inspection, and manufacturing architecture.
- If direct skin contact or sterilization is introduced, material selection must include biocompatibility and cleaning requirements not evaluated here.

9. Decision

Selected Baseline Material

6061-T6 aluminum remains the selected baseline for the side plates, frames, and spacers because it achieved the highest weighted score for this educational prototype. The selection is conditional on the current static load case, machining-oriented build plan, and non-clinical scope.

10. Verification Actions

Action	Purpose	Status
Confirm CAD material assignments	Ensure mass and FEA use intended baseline properties	Completed for analysis baseline
Obtain stock certifications before fabrication	Confirm actual alloy and temper	Future
Inspect dimensions and finish	Confirm manufactured parts meet drawings	Future
Conduct static and cyclic testing	Compare physical response with analysis	Future
Reassess shaft material	Address wear and bearing-interface durability	Required before physical V2

References

Project TITAN Engineering Design Report, July 2026.

Kaiser Aluminum, flat-rolled aluminum product information for 6061 and 7075 alloys.

TIMET, TIMETAL 6-4 technical data sheet.

Typical candidate-property ranges are used for conceptual comparison; procurement must use supplier-certified values.